

Clinical Reasoning & Explainability Report

Patient: Freddy Little | Cycle #3

1. Patient Context

Cancer Type: COLORECTAL
Age: 87 | Gender: M
Chemotherapy Drugs: Leucovorin, Oxaliplatin

2. Clinical Data Inputs & Their Impact

2a. Laboratory Values

Lab Test	Value	Normal	Status	Impact on Side Effects
WBC	2.7 x1000/mm3	4.5-11.0	LOW	Increased fever/infection risk and fatigue severity
Hemoglobin	12.6 g/dL	12-18	NORMAL	No modifier applied
GFR	12.0 mL/min	>60	LOW	Increased nausea/fatigue (drug toxicity accumulation)
Pain Score	3/10	0-3	LOW	No modifier applied

2b. Wearable Biometric Data

Metric	Value	Normal	Status	Impact
HRV (RMSSD)	22.8 ms	>30 ms	LOW	Autonomic stress -> increased fatigue, anxiety, SOB
Sleep Efficiency	92%	>85%	NORMAL	Adequate sleep quality
Mean Heart Rate	76 bpm	60-100	INFO	Contextual cardiovascular data

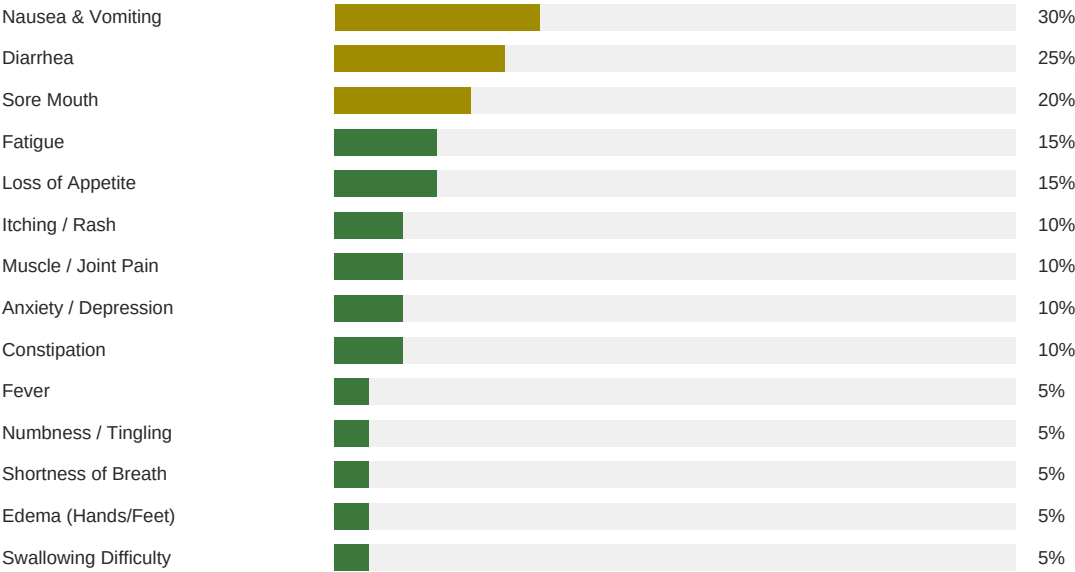
2c. Age Modifier

Age 87: 20% severity increase applied (elderly patients have reduced drug clearance)

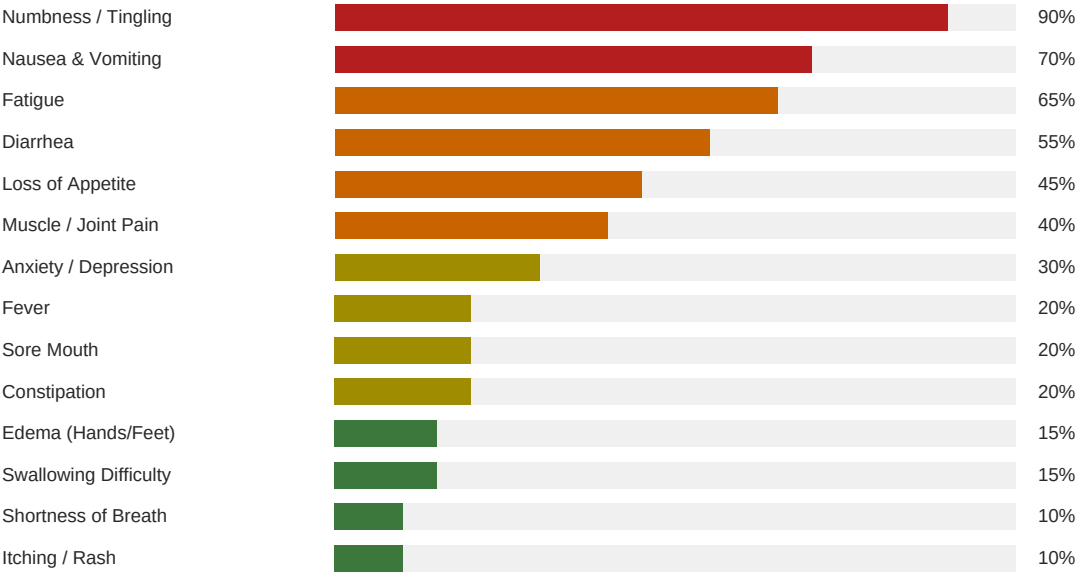
3. Drug Side-Effect Profiles

Base severity weights for each drug (0-1 scale, higher = more likely/severe)

Leucovorin -- Known Side Effect Affinities



Oxaliplatin -- Known Side Effect Affinities



4. Temporal Severity Model

Side-effect severity follows a temporal curve across the 7-day chemo cycle. Day 1 (infusion) starts mild, severity peaks at Days 3-5 (nadir period when blood counts are lowest), then eases on Days 6-7 (recovery). This pattern is well-established in oncology literature.

Day	Multiplier	Phase
Day 1	0.4x	Infusion
Day 2	0.7x	Onset
Day 3	1.0x	Nadir (Peak)
Day 4	1.0x	Nadir (Peak)
Day 5	0.9x	Late Nadir
Day 6	0.6x	Recovery
Day 7	0.4x	Recovery

5. Per-Side-Effect Severity Reasoning

Final Grade = Drug Base x Clinical Modifier x Temporal Curve x Cycle Factor + Random Noise

Fever

None	Mild	Mild	Mild	Mild	Mild	None
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Leucovorin base affinity: 5% | Oxaliplatin base affinity: 20% | Low WBC (2.7) amplifies this effect

Fatigue

Mode	Seve	Seve	Seve	Seve	Seve	Mode
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Leucovorin base affinity: 15% | Oxaliplatin base affinity: 65% | Low WBC (2.7) amplifies this effect | Low HRV (22.8ms) indicates aut

Nausea & Vomiting

Mild	Seve	Seve	Seve	Seve	Seve	Mild
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Leucovorin base affinity: 30% | Oxaliplatin base affinity: 70%

Sore Mouth

None	Mild	None	Mild	Mild	None	None
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Leucovorin base affinity: 20% | Oxaliplatin base affinity: 20%

Diarrhea

Mild	Mild	Mode	Mode	Mode	Mild	Mild
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Leucovorin base affinity: 25% | Oxaliplatin base affinity: 55%

Constipation

None	None	Mild	Mild	None	None	None
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Leucovorin base affinity: 10% | Oxaliplatin base affinity: 20%

Loss of Appetite

Mild	Mode	Mode	Mode	Mode	Mild	Mild
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Leucovorin base affinity: 15% | Oxaliplatin base affinity: 45%

Swallowing Difficulty

None	None	None	None	Mild	Mild	None
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Leucovorin base affinity: 5% | Oxaliplatin base affinity: 15%

Anxiety / Depression

None	Mild	Mild	Mode	None	None	None
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Leucovorin base affinity: 10% | Oxaliplatin base affinity: 30% | Low HRV (22.8ms) indicates autonomic stress

Edema (Hands/Feet)

None	None	None	None	Mild	None	None
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Leucovorin base affinity: 5% | Oxaliplatin base affinity: 15%

Itching / Rash

None	None	None	None	None	None	None
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Leucovorin base affinity: 10% | Oxaliplatin base affinity: 10%

Shortness of Breath

None	None	None	None	None	None	None
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Leucovorin base affinity: 5% | Oxaliplatin base affinity: 10% | Low HRV (22.8ms) indicates autonomic stress

Muscle / Joint Pain

Mild	Mild	Mild	Mode	Mode	None	None
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Leucovorin base affinity: 10% | Oxaliplatin base affinity: 40%

Numbness / Tingling

Mild	Seve	Seve	Seve	Seve	Mode	Mode
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Leucovorin base affinity: 5% | Oxaliplatin base affinity: 90%

6. Methodology & Data Sources

- SEVERITY CALCULATION: $\text{Final severity} = \text{Drug_Base_Weight} \times \text{Clinical_Modifier} \times \text{Temporal_Curve} \times \text{Cycle_Factor} + \text{Gaussian_Noise}(0, 0.08)$
- DRUG PROFILES: Based on published chemotherapy side-effect incidence rates from oncology literature.
- CLINICAL MODIFIERS: Lab values (WBC, Hemoglobin, GFR, Pain) adjust severity multipliers based on clinical thresholds.
- WEARABLE DATA: HRV (RMSSD) and sleep efficiency provide real-time physiological stress indicators.
- TEMPORAL MODEL: 7-day cycle follows established nadir pattern (peak toxicity Days 3-5 post-infusion).
- CYCLE FACTOR: Later cycles accumulate toxicity at +2% per cycle (capped at 40% increase).
- GRADING: Continuous 0-1 score mapped to 4 grades: None (<0.20), Mild (0.20-0.44), Moderate (0.45-0.69), Severe (>=0.70).
- DATA SOURCES: Synthea synthetic patient data (demographics, conditions, medications, observations, encounters) + generated wearable biometrics.